

DAY 6 INTRODUCTION TO NETWORK DEVICES THEORY

Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

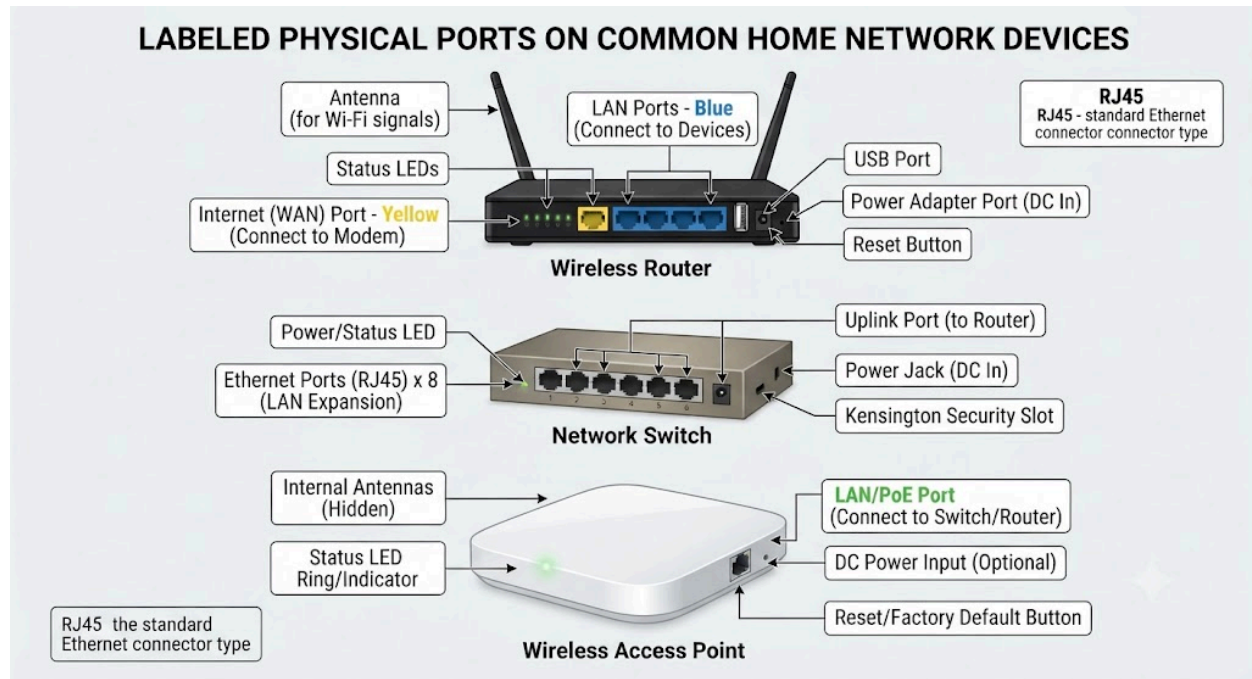
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Here is a table outlining the main functions and typical management or operational ports used by core networking devices in a home Wi-Fi setup:

Network Device	Main Function	Typical Ports Used
Router	Connects your home network to the internet (WAN) and routes traffic between different subnets (LAN).	Port 53 (DNS forwarding), Port 67/68 (DHCP), Port 80/443 (Web UI management)
Switch	Connects multiple wired devices together locally within the same network segment using MAC addresses.	Mostly Layer 2 (operates below port level), but managed switches use Port 22 (SSH) or Port 80/443 (Web UI)
Access Point (AP)	Converts wired network signals into wireless radio frequencies so Wi-Fi devices can connect.	Port 53 (DNS), Port 443 (Cloud/Controller management communication)
Firewall	Inspects incoming and outgoing network traffic and blocks unauthorized access based on security rules.	Varies by rule, but typically secures Port 443 (HTTPS) and filters inbound connection requests
Hub	(Legacy) Broadcasts incoming data packets to all ports blindly without filtering or routing intelligence.	<i>None</i> (Operates purely at Layer 1 / Physical layer with no IP or port awareness)

Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

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Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

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When you open Zomato to order food on Wi-Fi, your phone first connects to a wireless **access point** to convert your radio signals into wired data. Next, your home **router** receives that packet, applies Network Address Translation (NAT), and forwards your food order across the internet to Zomato's servers. Finally, an enterprise **firewall** on the server side inspects the incoming connection request to ensure secure and safe data delivery.

Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Hint: Think about device capacity, coverage, and traffic handling.

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Possible Reason: Too many students connecting simultaneously overwhelms the wireless **Access Point's** client capacity, leading to bandwidth congestion, high latency, and dropped connections.

How Upgrading Helps: Upgrading to modern, enterprise-grade **Access Points** (with Wi-Fi 6 or higher and features like MU-MIMO) can handle hundreds of concurrent device connections, distribute traffic efficiently across channels, and eliminate random disconnects.